

# Park 2016 Table-4 OR vs mean |SHAP| rank (migraine/park)

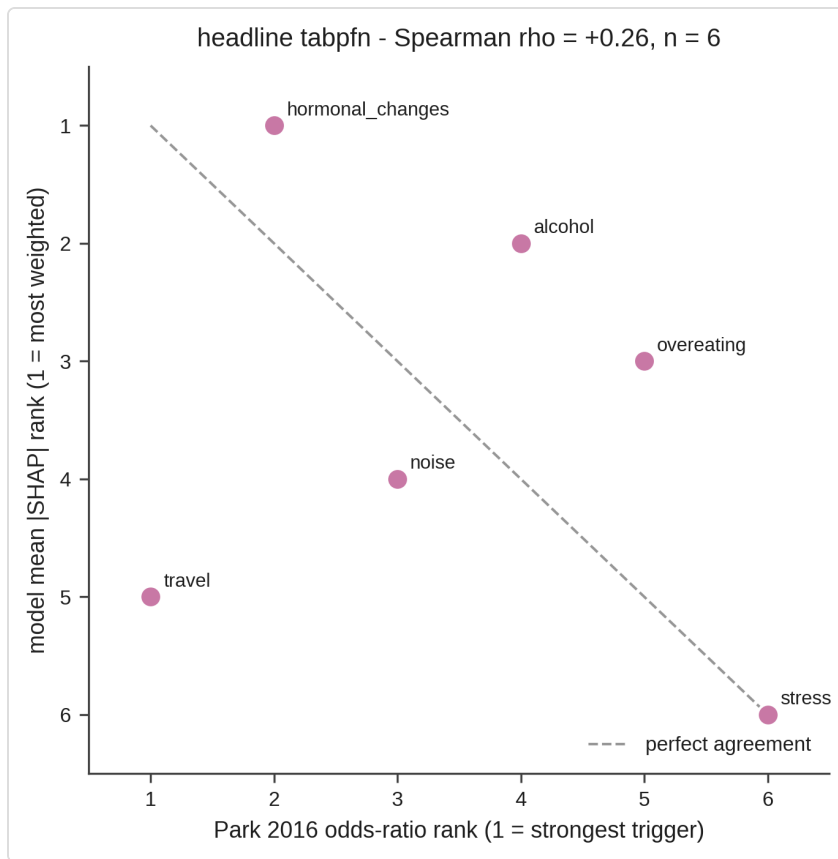
**What this compares.** Park et al. 2016 report odds ratios for self-reported migraine triggers in the SHD cohort - a population-level, same-day association between a trigger and a migraine day. The model's mean |SHAP| rank is a data-driven importance for *next-day* individual prediction from the park trigger features. The two answer different questions, so this is a convergence check, not a validation: a positive Spearman correlation means the model recovers Park's trigger ordering; a value near zero or negative means it weights the triggers differently, which is expected when same-day cross-sectional odds ratios are asked to drive a next-day forecast on a small feature set. Each panel shows the leaf's full provenance (architecture, hyperparameter-tuning, data-split ratio and split type) and the odds-ratio-rank vs SHAP-rank scatter against the agreement diagonal.

**headline: tabpfn (tabpfn, mean |SHAP|)**

**tabpfn/version\_3-binary** [NonHP (library defaults)] | split 70\_30/chrono | hold-out AUROC 0.646, AUPRC 0.126, calib +1.05

Spearman correlation (SHAP rank vs Park OR rank): **+0.257 (p = 0.623, n = 6)** over 6 triggers.

| trigger                | Park OR | OR rank | SHAP rank | mean  SHAP |
|------------------------|---------|---------|-----------|------------|
| travel_today           | 6.4     | 1       | 5         | 0.0153     |
| hormonal_changes_today | 3.5     | 2       | 1         | 0.0532     |
| noise_today            | 2.8     | 3       | 4         | 0.0194     |
| alcohol_today          | 2.5     | 4       | 2         | 0.0228     |
| overeating_today       | 2.4     | 5       | 3         | 0.0219     |
| stress_today           | 1.8     | 6       | 6         | 0.0151     |

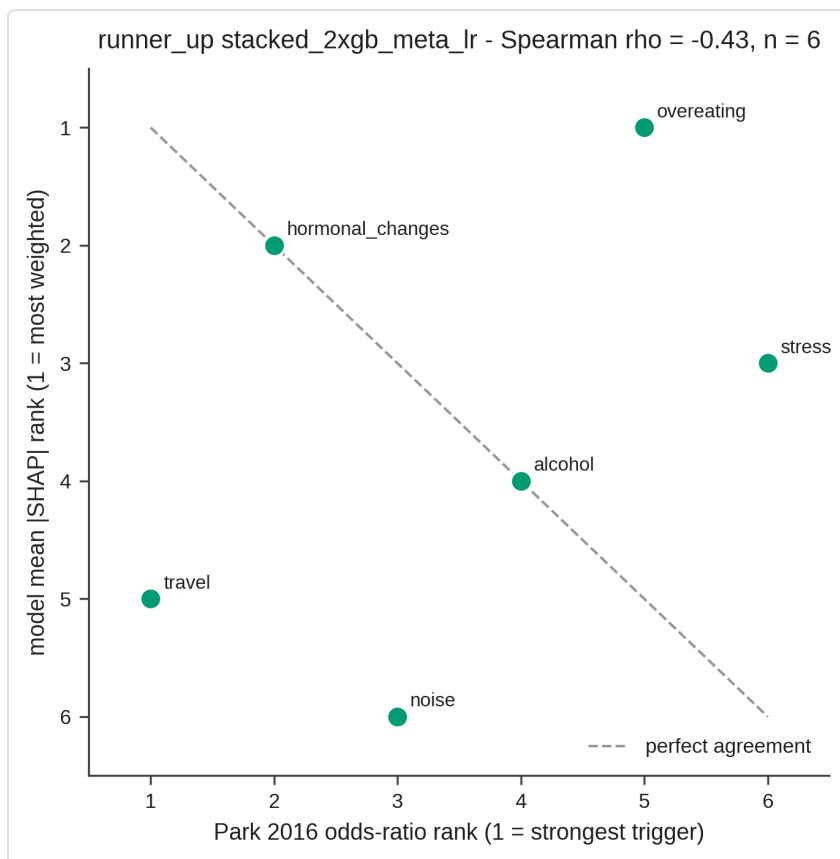


**runner\_up: stacked\_2xgb\_meta\_lr (xgboost, mean |SHAP|)**

**stacked\_2xgb\_meta\_lr** [NonHP (library defaults)] | split 70\_30/chrono | hold-out AUROC 0.625, AUPRC 0.115, calib +0.82

Spearman correlation (SHAP rank vs Park OR rank): **-0.429 (p = 0.397, n = 6)** over 6 triggers.

| trigger                | Park OR | OR rank | SHAP rank | mean  SHAP |
|------------------------|---------|---------|-----------|------------|
| travel_today           | 6.4     | 1       | 5         | 0.0088     |
| hormonal_changes_today | 3.5     | 2       | 2         | 0.0279     |
| noise_today            | 2.8     | 3       | 6         | 0.0054     |
| alcohol_today          | 2.5     | 4       | 4         | 0.0157     |
| overeating_today       | 2.4     | 5       | 1         | 0.0292     |
| stress_today           | 1.8     | 6       | 3         | 0.0205     |

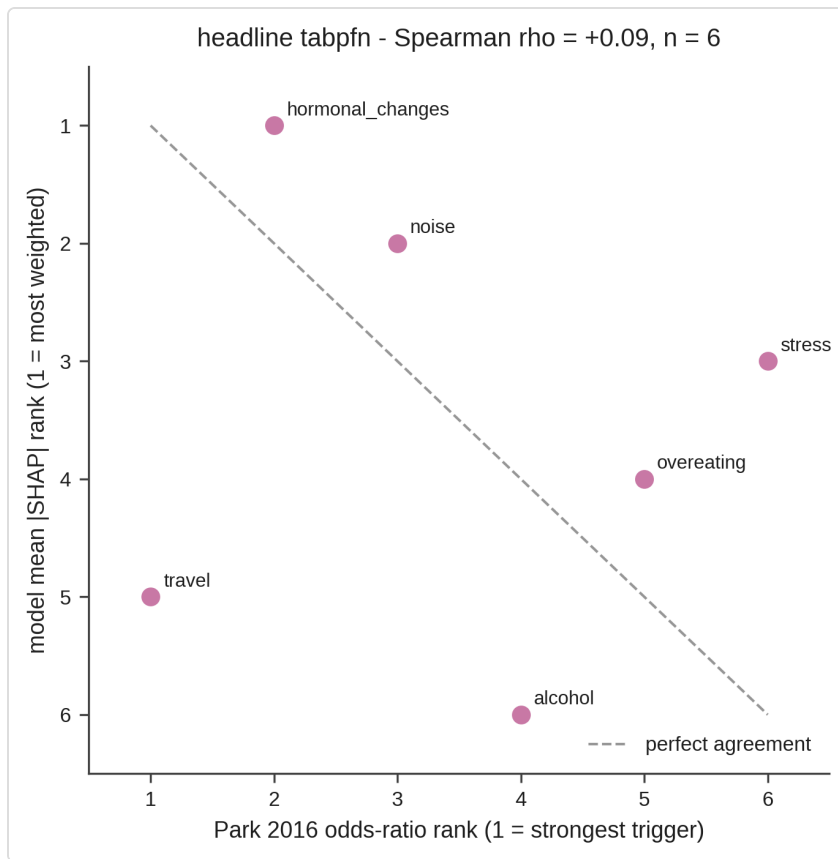


## headline: tabpfn (tabpfn, mean |SHAP|)

**tabpfn/version\_3-default** [NonHP (library defaults)] | split 80\_20/stratified | hold-out AUROC 0.618, AUPRC 0.119, calib +0.59

Spearman correlation (SHAP rank vs Park OR rank): **+0.086 (p = 0.872, n = 6)** over 6 triggers.

| trigger                | Park OR | OR rank | SHAP rank | mean  SHAP |
|------------------------|---------|---------|-----------|------------|
| travel_today           | 6.4     | 1       | 5         | 0.0045     |
| hormonal_changes_today | 3.5     | 2       | 1         | 0.0136     |
| noise_today            | 2.8     | 3       | 2         | 0.0099     |
| alcohol_today          | 2.5     | 4       | 6         | 0.0034     |
| overeating_today       | 2.4     | 5       | 4         | 0.0057     |
| stress_today           | 1.8     | 6       | 3         | 0.0075     |

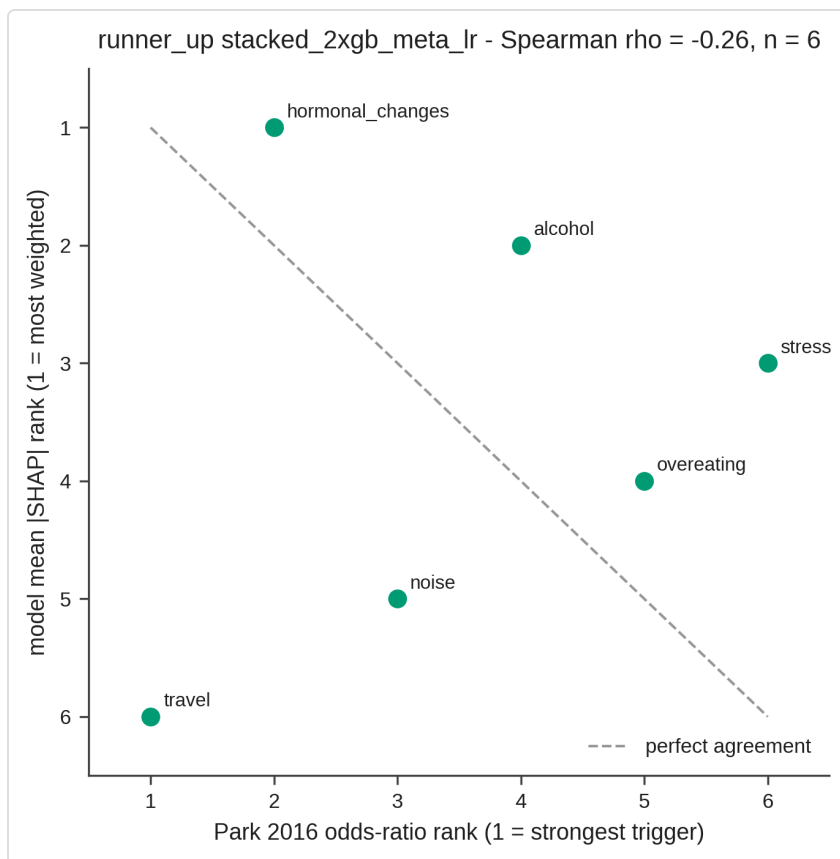


**runner\_up: stacked\_2xgb\_meta\_lr (xgboost, mean |SHAP|)**

**stacked\_2xgb\_meta\_lr** [NonHP (library defaults)] | split 70\_30/stratified | hold-out AUROC 0.576, AUPRC 0.117, calib +0.56

Spearman correlation (SHAP rank vs Park OR rank): **-0.257 (p = 0.623, n = 6)** over 6 triggers.

| trigger                | Park OR | OR rank | SHAP rank | mean  SHAP |
|------------------------|---------|---------|-----------|------------|
| travel_today           | 6.4     | 1       | 6         | 0.0012     |
| hormonal_changes_today | 3.5     | 2       | 1         | 0.0338     |
| noise_today            | 2.8     | 3       | 5         | 0.0023     |
| alcohol_today          | 2.5     | 4       | 2         | 0.0115     |
| overeating_today       | 2.4     | 5       | 4         | 0.0043     |
| stress_today           | 1.8     | 6       | 3         | 0.0095     |

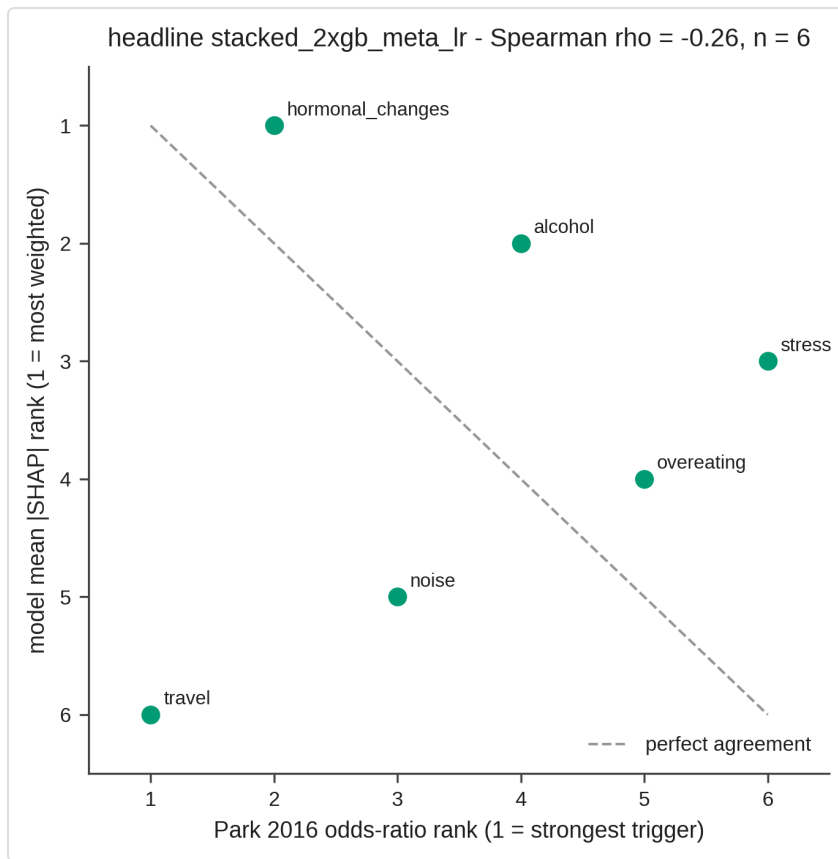


## headline: stacked\_2xgb\_meta\_lr (xgboost, mean |SHAP|)

**stacked\_2xgb\_meta\_lr** [NonHP (library defaults)] | split 70\_30/patient | hold-out AUROC 0.541, AUPRC 0.086, calib +0.67

Spearman correlation (SHAP rank vs Park OR rank): **-0.257 (p = 0.623, n = 6)** over 6 triggers.

| trigger                | Park OR | OR rank | SHAP rank | mean  SHAP |
|------------------------|---------|---------|-----------|------------|
| travel_today           | 6.4     | 1       | 6         | 0.0000     |
| hormonal_changes_today | 3.5     | 2       | 1         | 0.0001     |
| noise_today            | 2.8     | 3       | 5         | 0.0000     |
| alcohol_today          | 2.5     | 4       | 2         | 0.0001     |
| overeating_today       | 2.4     | 5       | 4         | 0.0000     |
| stress_today           | 1.8     | 6       | 3         | 0.0000     |



**runner\_up: tabpfn (tabpfn, mean |SHAP|)**

**tabpfn/version\_3-default** [NonHP (library defaults)] | split 70\_30/patient | hold-out AUROC 0.536, AUPRC 0.082, calib +0.18

Spearman correlation (SHAP rank vs Park OR rank): **-0.029 (p = 0.957, n = 6)** over 6 triggers.

| trigger                | Park OR | OR rank | SHAP rank | mean  SHAP |
|------------------------|---------|---------|-----------|------------|
| travel_today           | 6.4     | 1       | 6         | 0.0055     |
| hormonal_changes_today | 3.5     | 2       | 1         | 0.0220     |
| noise_today            | 2.8     | 3       | 4         | 0.0081     |
| alcohol_today          | 2.5     | 4       | 2         | 0.0087     |
| overeating_today       | 2.4     | 5       | 3         | 0.0085     |
| stress_today           | 1.8     | 6       | 5         | 0.0060     |

runner\_up tabpfn - Spearman rho = -0.03, n = 6

